8 Annex B

(normative)

Process Reference Model (PRM) for Assessment Purposes

B.1 Introduction

It is understood that some users of this International Standard may desire to assess the implemented processes in accordance with ISO/IEC 15504-2, *Information Technology — Process Assessment — Part 2: Performing an Assessment*.

This Annex provides a Process Reference Model (PRM) suitable for use in conjunction with ISO/IEC 15504-2.

The source for the processes in this model is the set of processes defined in the body of this International Standard. In each case, the name, statement of purpose, and statement of outcomes for each process in the body of this standard have been referenced for use in this Annex. In some cases, the processes in the body of the standard have been explicitly identified as lower-level processes or replacement processes for equivalent clauses in ISO/IEC 15288.

B.2 Conformance with ISO/IEC 15504-2

B.2.1 General

The Process Reference Model included in this Annex is suitable for use in process assessments performed in accordance with ISO/IEC 15504-2, *Information Technology — Process Assessment — Part 2: Performing an Assessment*.

Each of the following clauses includes italicized text quoting the requirements from ISO/IEC 15504-2, followed by upright text describing how these requirements are satisfied by this International Standard.

B.2.2 Requirements for Process Reference Models

A Process Reference Model shall contain:

a) **A declaration of the domain of the Process Reference Model.**
This is provided in Clause 1 of this International Standard.

b) **A description of the processes within the scope of the Process Reference Model.**
This is provided in Annex B.3, meeting the requirements of subclause 6.2.4 of this International Standard.

c) **A description of the relationship between the Process Reference Model and its intended context of use.**
This is provided in Clause 5 of this International Standard.

d) **A description of the relationship between the processes defined within the Process Reference Model.**
This is provided in Annex B.3 in the description of each process. For example, some process descriptions in this Annex include statements that the process is a lower-level process and that it replaces equivalent clauses in ISO/IEC 15288.

The Process Reference Model shall document the community of interest of the model and the actions taken to achieve consensus within that community of interest:

a) **The relevant community of interest shall be characterized or specified.**
The relevant community of interest comprises the users of ISO/IEC 15288 and ISO/IEC 12207.

b) **The extent of achievement of consensus shall be documented.**
Both ISO/IEC 15288 and ISO/IEC 12207 are International Standards developed and approved through the ISO/IEC JTC 1 process, which satisfies the formal consensus requirements.

c) **If no actions are taken to achieve consensus, a statement to this effect shall be documented.**
This is not applicable, as the required consensus actions have been undertaken through the international standardization process.

The processes defined within a Process Reference Model shall have **unique**

process descriptions and identification.
The process descriptions provided in this Annex are unique. Identification is ensured by unique process names and by the clause numbering of this Annex.

B.2.3 Process Descriptions

The fundamental elements of a Process Reference Model are the descriptions of the processes within the scope of the model.

The process descriptions in this PRM incorporate:

- a **statement of purpose** for each process, which describes at a high level the overall objectives of performing the process; and
- a **set of outcomes**, which demonstrate successful achievement of the process purpose.

These process descriptions shall meet the following requirements:

a) A process shall be described in terms of its purpose and outcomes.
b) In any process description, the set of process outcomes shall be **necessary and sufficient** to achieve the purpose of the process.
c) Process descriptions shall be such that **no aspects of the Measurement Framework** as described in Clause 5 of ISO/IEC 15504-2 beyond **Capability Level 1** are contained or implied.

An **outcome statement** describes one of the following:

- the production of an artefact;
- a significant change of state; or
- the meeting of specified constraints (e.g., requirements, goals, conditions).

These requirements are met by the process descriptions in this Annex. Some outcomes might be interpreted as contributing to levels of capability above Level 1; however, a conforming implementation of the relevant processes does not require achievement of these higher levels of capability.

B.2.4 Common Process Attributes for Capability Determination

The attributes described in 5.1.9 of this International Standard characterize the specificity of each process. When an implemented process conforms to these attributes, the process' defined purpose and outcomes are achieved through the implementation of its defined activities.

In addition to these basic attributes, processes may be characterized by **other attributes common to all processes**. These common attributes contribute to the achievement of higher levels of process capability as defined in ISO/IEC 15504-2.

There are six levels of process capability in the Measurement Framework of ISO/IEC 15504-2, as described in the following table:

Table B.1 — Six Levels of Process Capability

Capability Level	Process Capability
0	Incomplete Process
1	Performed Process
2	Managed Process
3	Established Process
4	Predictable Process
5	Optimizing Process

B.2.5 Common Process Attributes for Higher Capability Levels

The achievement of higher-level attributes and capabilities is enabled by the interaction of the process with support and organizational processes such as **Documentation**, **Configuration Management**, **Quality Assurance**, **Measurement**, and **Improvement**.

ISO/IEC 15504-2 identifies the following **common process attributes (PA)** affiliated with the achievement of higher levels of process capability:

- **Performance Management (PA 2.1)** — Determines the extent to which the performance of the process is managed.
The achievement of this attribute involves planning, monitoring, and adjusting the process performance.
- **Work Product Management (PA 2.2)** — Determines the extent to which the work products produced by the process are appropriately managed.
The achievement of this attribute ensures that work products are properly established, controlled, and maintained.
- **Process Definition (PA 3.1)** — Determines the extent to which the

process is established as a **standard process** within the organization.
The achievement of this attribute involves the definition of the process in terms of:

- required competencies and roles for performing the process;
- required infrastructure and work environment;
- methods for monitoring its effectiveness and suitability; and
- tailoring guidelines.
- **Process Deployment (PA 3.2)** — Determines the extent to which the process is effectively deployed as a tailored instance of the standard process.

The achievement of this attribute is reflected in fidelity to the standard process, effective deployment of resources to the implementation of the process, and the collection and analysis of data for understanding and refining process behaviour.

- **Process Measurement (PA 4.1)** — Determines the extent to which process measurements are used to ensure that process performance supports the achievement of defined business goals.

The achievement of this attribute is concerned with the existence of an effective system for the **collection of measures** relevant to the performance of the process and the quality of the work products. The measures are applied to determine the extent of achievement of the organization's business goals.

- **Process Control (PA 4.2)** — Determines the extent to which the process is quantitatively managed to produce a process that is stable, capable, and predictable within defined limits.

The achievement of this attribute implies the application of analysis and control techniques to ensure that the process performs within defined limits and that corrective actions are taken to address deviations.

- **Process Innovation (PA 5.1)** — Determines the extent to which changes to the process are identified from analysis of variation in performance and from investigations of innovative approaches to process definition and implementation.

The achievement of this attribute is concerned with the existence of a **proactive focus on continuous improvement** to fulfil both current and projected business goals.

- **Process Optimization (PA 5.2)** — Determines the extent to which changes to the definition, management, and performance of the process result in effective impact that achieves relevant process improvement objectives.

The achievement of this attribute is concerned with an **orderly and proactive approach** to identifying and introducing appropriate changes to the process, minimizing undesired disruption, and evaluating their effects to ensure improvements are realized.

B.3 Process Reference Model

The **Process Reference Model** is composed of the **statement of purpose** and **set of outcomes** for each of the processes included in **Clause 6** and **Clause 7** of this International Standard.

These processes are organized by their life cycle categories (e.g., Agreement, Organizational, Project, Technical, and Support), and each process description includes:

- the **process name**;
- the **process purpose statement**; and
- the **process outcomes**, which collectively provide a **necessary and sufficient** set to achieve the process purpose.

The set of processes is listed in **Table B.2**.

Table B.2 — ISO/IEC 12207:2008 Processes

ISO/IEC 12207 Clause Number	ISO/IEC 12207:2008 Process Name
6	*System Life Cycle Processes*
6.1	*Agreement Processes*
6.1.1	*Acquisition Process*
6.1.2	*Supply Process*
6.2	*Organizational Project-Enabling Processes*
6.2.1	*Life Cycle Model Management Process*
6.2.2	*Infrastructure Management Process*
6.2.3	*Project Portfolio Management Process*
6.2.4	*Human Resource Management Process*
6.2.5	*Quality Management Process*
6.3	*Project Processes*
6.3.1	*Project Planning Process*
6.3.2	*Project Assessment and Control Process*
6.3.3	*Decision Management Process*
6.3.4	*Risk Management Process*
6.3.5	*Configuration Management Process*
6.3.6	*Information Management Process*
6.3.7	*Measurement Process*
6.4	*Technical Processes*
6.4.1	*Stakeholder Requirements Definition Process*
6.4.2	*System Requirements Analysis*
6.4.3	*System Architectural Design*
6.4.4	*Implementation Process*
6.4.5	*System Integration Process*
6.4.6	*System Qualification Testing Process*
6.4.7	*Software Installation*
6.4.8	*Software Acceptance Support*
6.4.9	*Software Operation Process*
6.4.10	*Software Maintenance Process*
6.4.11	*Software Disposal Process*
7	*Software Life Cycle Processes*
7.1	*Software Implementation Processes*
7.1.1	*Software Implementation Process*
7.1.2	*Software Requirements Analysis Process*
7.1.3	*Software Architectural Design Process*
7.1.4	*Software Detailed Design Process*
7.1.5	*Software Construction Process*
7.1.6	*Software Integration Process*

ISO/IEC 12207 Clause Number	ISO/IEC 12207:2008 Process Name
7.1.7	*Software Qualification Testing Process*
7.2	*Software Support Processes*
7.2.1	*Software Documentation Management Process*
7.2.2	*Software Configuration Management Process*
7.2.3	*Software Quality Assurance Process*
7.2.4	*Software Verification Process*
7.2.5	*Software Validation Process*
7.2.6	*Software Review Process*
7.2.7	*Software Audit Process*
7.2.8	*Software Problem Resolution Process*
7.3	*Software Reuse Processes*
7.3.1	*Domain Engineering Process*
7.3.2	*Reuse Asset Management Process*
7.3.3	*Reuse Program Management Process*

NOTE — These process descriptions provide the basis for process assessment at **Capability Level 1** in accordance with ISO/IEC 15504-2. Higher levels of capability are evaluated by applying the common process attributes defined in Clause B.2.5 in conjunction with this PRM.

Some activities of the processes in Clauses 6 and 7 are replaced with corresponding **lower-level processes**. The descriptions of these lower-level processes are provided below.

B.3.1 Acquisition Process Lower-Level Processes

B.3.1.1 Acquisition Preparation Process

This process is a lower-level process of the Acquisition Process. It replaces the *Acquisition Preparation* activity (6.1.1.3.1).

B.3.1.1.1 Purpose
The purpose of the Acquisition Preparation Process is to establish the needs and goals of the acquisition and to communicate these with potential suppliers.

B.3.1.1.2 Outcomes

As a result of successful implementation of the Acquisition Preparation Process:
a) the concept or the need for the acquisition, development, or enhancement is established;
b) stakeholder requirements are defined;
c) an acquisition strategy is developed; and
d) supplier selection criteria are defined.

B.3.1.2 Supplier Selection Process

This process is a lower-level process of the Acquisition Process. It replaces the *Supplier Selection* activity (6.1.1.3.3).

B.3.1.2.1 Purpose

The purpose of the Supplier Selection Process is to choose the organization that is to be responsible for the delivery of the project requirements.

B.3.1.2.2 Outcomes

As a result of successful implementation of the Supplier Selection Process:
a) the supplier selection criteria are established and used to evaluate potential suppliers;
b) the supplier is selected based on evaluation of proposals, process capabilities, and other relevant factors; and
c) an agreement is established and negotiated between the acquirer and the supplier.

B.3.1.3 Agreement Monitoring Process

This process is a lower-level process of the Acquisition Process. It replaces the *Agreement Monitoring* activity (6.1.1.3.5).

B.3.1.3.1 Purpose

The purpose of the Agreement Monitoring Process is to track and assess the performance of the supplier against agreed requirements.

B.3.1.3.2 Outcomes

As a result of successful implementation of the Agreement Monitoring Process:
a) joint activities between the acquirer and the supplier are performed as needed;
b) information on technical progress is exchanged regularly with the supplier;
c) supplier performance is monitored against agreed requirements; and
d) agreement changes, if needed, are negotiated between the acquirer and supplier and documented.

B.3.1.4 Acquirer Acceptance Process

This process is a lower-level process of the Acquisition Process. It replaces the *Acquirer Acceptance* activity (6.1.1.3.6).

B.3.1.4.1 Purpose

The purpose of the Acquirer Acceptance Process is to approve the supplier's deliverable when all acceptance criteria are satisfied.

B.3.1.4.2 Outcomes

As a result of successful implementation of the Acquirer Acceptance Process:
a) the delivered software product and/or service is evaluated with respect to the agreement;
b) the acquirer's acceptance is based on the agreed acceptance criteria; and
c) the software product and/or service is accepted by the acquirer.

B.3.2 Supply Process Lower-Level Processes

B.3.2.1 Supplier Tendering Process

This process is a lower-level process of the Supply Process. It replaces the *Supplier Tendering* activity (6.1.2.3.2).

B.3.2.1.1 Purpose

The purpose of the Supplier Tendering Process is to establish an interface to respond to acquirer inquiries and requests for proposal, and to prepare and submit proposals.

B.3.2.1.2 Outcomes

As a result of successful implementation of the Supplier Tendering Process:
a) a communication interface is established and maintained to respond to acquirer inquiries and RFPs;
b) requests for proposal are evaluated to determine whether to submit a proposal;
c) the need for preliminary surveys or feasibility studies is determined;
d) suitable resources are identified to perform the proposed work; and
e) a supplier proposal is prepared and submitted.

B.3.2.2 Contract Agreement Process

This process is a lower-level process of the Supply Process. It replaces the *Contract Agreement* activity (6.1.2.3.4).

B.3.2.2.1 Purpose

The purpose of the Contract Agreement Process is to negotiate and approve a contract/agreement that clearly specifies expectations, responsibilities, deliverables, and liabilities of both the supplier and the acquirer.

B.3.2.2.2 Outcomes

As a result of successful implementation of the Contract Agreement Process:
a) a contract/agreement is negotiated, reviewed, approved, and awarded;
b) mechanisms for monitoring supplier performance and mitigating risks are reviewed and considered for inclusion;
c) proposers/tenderers are notified of selection results; and
d) formal confirmation of agreement is obtained.

The Contract Agreement Process is used to obtain formal confirmation of assignments offered during the Supplier Tendering Process.

B.3.2.3 Product/Service Delivery and Support Process

This process is a lower-level process of the Supply Process. It replaces the *Product/Service Delivery and Support* activity (6.1.2.3.6).

B.3.2.3.1 Purpose

The purpose of the Product/Service Delivery and Support Process is to provide the specified product or service to the acquirer with appropriate support to ensure that the requirements have been met.

B.3.2.3.2 Outcomes

As a result of successful implementation of the Product/Service Delivery and Support Process:
a) the contents of the product release are determined;
b) the release is assembled from configured items;
c) the release documentation is produced;
d) the delivery mechanism and media are determined;
e) release approval is effected against defined criteria;
f) the product release is made available to the acquirer;
g) confirmation of release is obtained;

h) the product is delivered to the acquirer;
i) acquirer acceptance tests and reviews are supported;
j) the product is put into operation in the customer’s environment; and
k) problems detected during acceptance are identified and communicated for resolution.

NOTE — Incremental delivery may be performed in completed increments.

B.3.3 Life Cycle Model Management Process Lower-Level Processes

B.3.3.1 Process Establishment Process

This process is a lower-level process of the Life Cycle Model Management Process. It replaces the *Process Establishment* activity (6.2.1.3.1).

B.3.3.1.1 Purpose
The purpose of the Process Establishment Process is to establish a suite of organizational processes for all life cycle processes as they apply to business activities.

B.3.3.1.2 Outcomes
As a result of successful implementation of the Process Establishment Process:
a) a defined and maintained standard set of processes is established, with indications of applicability;
b) the detailed tasks, activities, and work products of the standard process are identified, with expected performance characteristics;
c) a strategy for tailoring and applying the standard process to products or services is developed; and
d) information and data related to use of the standard process for specific projects are maintained.

B.3.3.2 Process Assessment Process

This process is a lower-level process of the Life Cycle Model Management Process. It replaces the *Process Assessment* activity (6.2.1.3.2).

B.3.3.2.1 Purpose
The purpose of the Process Assessment Process is to determine the extent of process implementation and identify opportunities for continuous improvement.

B.3.3.2.2 Outcomes
As a result of successful implementation of the Process Assessment Process:
a) information and data related to the use of the standard process for specific projects exist and are maintained;
b) the strengths and weaknesses of the organization's standard processes are understood; and
c) accurate and accessible assessment records are maintained.

B.3.3.3 Process Improvement Process

This process is a lower-level process of the Life Cycle Model Management Process. It replaces the *Process Improvement* activity (6.2.1.3.3).

B.3.3.3.1 Purpose
The purpose of the Process Improvement Process is to continually improve organizational effectiveness and efficiency through improvements to processes aligned with business needs.

B.3.3.3.2 Outcomes
As a result of successful implementation of the Process Improvement Process:
a) commitment is established to provide resources to sustain improvement actions;
b) internal and external issues are identified as improvement opportunities;
c) the current status of processes is analyzed to identify improvement stimuli;
d) improvement goals are identified and prioritized, and changes are implemented;
e) the effects of improvements are monitored and confirmed against goals;
f) knowledge gained from improvements is communicated throughout the organization; and
g) improvements are evaluated for reuse elsewhere in the organization.

NOTE 1 — Information sources include assessments, audits, customer feedback, cost of quality, and organizational performance reports.
NOTE 2 — Process assessment may be used to determine current process status.

B.3.4 Human Resource Management Process Lower-Level Processes

B.3.4.1 Skill Development Process

This process is a lower-level process of the Human Resource Management

Process. It replaces the *Skill Development* activity (6.2.4.3.2).

B.3.4.1.1 Purpose
The purpose of the Skill Development Process is to provide individuals with the skills and knowledge needed to perform their roles effectively.

B.3.4.1.2 Outcomes
As a result of successful implementation of the Skill Development Process:
a) training is developed or acquired to meet organizational and project needs; and
b) training is conducted to ensure individuals can perform their assignments effectively.

B.3.4.2 Skill Acquisition and Provision Process

This process is a lower-level process of the Human Resource Management Process. It replaces the *Skill Acquisition and Provision* activity (6.2.4.3.3).

B.3.4.2.1 Purpose
The purpose of the Skill Acquisition and Provision Process is to provide the organization and projects with individuals who possess the necessary skills and knowledge to work effectively as a cohesive group.

B.3.4.2.2 Outcomes
As a result of successful implementation of the Skill Acquisition and Provision Process:
a) individuals with the required competencies are identified and recruited;
b) effective interaction between individuals and groups is supported;
c) the workforce has the skills to share information and coordinate efficiently; and
d) objective performance criteria are defined and used for feedback and enhancement.

B.3.4.3 Knowledge Management Process

This process is a lower-level process of the Human Resource Management Process. It replaces the *Knowledge Management* activity (6.2.4.3.4).

B.3.4.3.1 Purpose
The purpose of the Knowledge Management Process is to ensure that knowledge, information, and skills are collected, shared, reused, and improved throughout the organization.

B.3.4.3.2 Outcomes
As a result of successful implementation of the Knowledge Management Process:
a) infrastructure is established and maintained for sharing information across the organization;
b) knowledge is readily available and shared; and
c) an appropriate knowledge management strategy is selected.

B.3.5 Software Operation Process Lower-Level Processes

B.3.5.1 Operational Use Process

This process is a lower-level process of the Software Operation Process. It replaces the *Operational Use* activity (6.4.9.3.3).

B.3.5.1.1 Purpose
The purpose of the Operational Use Process is to ensure the correct and efficient operation of the product during its intended usage.

B.3.5.1.2 Outcomes
As a result of successful implementation of the Operational Use Process:
a) operational risks for product introduction and operation are identified and monitored;
b) the product is operated in its intended environment according to requirements; and
c) criteria for operational use are developed to demonstrate compliance with agreed requirements.

B.3.5.2 Customer Support Process

This process is a lower-level process of the Software Operation Process. It replaces the *Customer Support* activity (6.4.9.3.4).

B.3.5.2.1 Purpose
The purpose of the Customer Support Process is to ensure that responsive and effective support is provided to customers during operational use.

B.3.5.2.2 Outcomes
As a result of successful implementation of the Customer Support Process:
a) service needs for customer support are identified and monitored;

b) customer satisfaction with support services and the product is evaluated on an ongoing basis;
c) operational support is provided by handling customer inquiries, requests, and resolving problems; and
d) customer support needs are met through appropriate services.